

STEP 1: Upload the dataset

```
from google.colab import files
```

```
uploaded = files.upload() # Upload 'fake_news_200_sample.csv'
```

STEP 2: Load and view data

```
import pandas as pd
```

```
df = pd.read_csv("fake_news_200_sample.csv")
```

```
print("Sample Data:\n")
```

```
print(df.head())
```

STEP 3: Preprocess the text

```
import re
```

```
import string
```

```
def clean_text(text):
```

```
    text = text.lower()
```

```
    text = re.sub(r'https?://\S+|www\.\S+', '', text)
```

```
    text = re.sub(r'<.*?>+', '', text)
```

```
    text = re.sub(r'\[.*?\]', '', text)
```

```
    text = re.sub(f"[{re.escape(string.punctuation)}]", '', text)
```

```
    text = re.sub(r'\n', '', text)
```

```
    return text
```

```
df['text'] = df['text'].apply(clean_text)
```

STEP 4: Train-Test Split and Vectorization

```
from sklearn.model_selection import train_test_split
```

```
from sklearn.feature_extraction.text import TfidfVectorizer
```

```
X = df['text']
y = df['label']
X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.2, random_state=42)
vectorizer = TfidfVectorizer(stop_words='english', max_df=0.7)
X_train_vec = vectorizer.fit_transform(X_train)
X_test_vec = vectorizer.transform(X_test)
```

STEP 5: Train the Model

```
from sklearn.linear_model import PassiveAggressiveClassifier
from sklearn.metrics import accuracy_score, confusion_matrix, classification_report
model = PassiveAggressiveClassifier(max_iter=50)
model.fit(X_train_vec, y_train)
y_pred = model.predict(X_test_vec)
accuracy = accuracy_score(y_test, y_pred)
print(f"\n✅ Model Accuracy: {accuracy * 100:.2f}%")
```

STEP 6: Predict on New News

```
def predict_news(text):
    cleaned = clean_text(text)
    vec = vectorizer.transform([cleaned])
    pred = model.predict(vec)[0]
    return "✅ REAL NEWS" if pred == 1 else "❌ FAKE NEWS"

# Test
sample_text = "Government announces a new scheme to support startups."
print("\nSample Prediction:", predict_news(sample_text))
```

STEP 7: Confusion Matrix Graph

```
import matplotlib.pyplot as plt
import seaborn as sns

cm = confusion_matrix(y_test, y_pred)

labels = ['FAKE', 'REAL']

plt.figure(figsize=(6,4))

sns.heatmap(cm, annot=True, fmt='d', cmap='Blues', xticklabels=labels,
yticklabels=labels)

plt.title("Confusion Matrix")

plt.xlabel("Predicted")

plt.ylabel("Actual")

plt.show()
```

STEP 8: Show Classification Report

```
report = classification_report(y_test, y_pred, target_names=labels)

print("\n📄 Classification Report:\n")

print(report)
```